

Vector Calculus Operators via the Nabla Symbol

The shortest and easiest way to write (and remember) the gradient, divergence, and curl uses the symbol ∇ (often typeset as $\vec{\nabla}$), a differential operator analogous to ∂_x , ∂_y , etc. In Cartesian coordinates it is defined as:

$$\vec{\nabla} \equiv \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

and is called either *del* or *nabla*. Acting with ∇ on different kinds of fields gives the standard vector calculus operators:

Gradient (scalar $\rightarrow$ vector):

$$\text{Gradient:} \quad \nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

Divergence (vector $\rightarrow$ scalar):

$$\text{Divergence:} \quad \nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

Curl (vector $\rightarrow$ vector):

$$\text{Curl:} \quad \nabla \times \mathbf{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix}$$

The Laplacian of a scalar field is

$$\nabla^2 f \equiv \nabla \cdot (\nabla f)$$

These operators appear naturally in fluid flow through porous media. For example, Darcy's law relates the volumetric flux (specific discharge) $\mathbf{q}(\mathbf{x})$ to the pressure gradient:

$$\mathbf{q} = -\frac{1}{\mu} \mathbf{K} (\nabla p - \rho \mathbf{g})$$

Here $p(\mathbf{x})$ is fluid pressure, ρ is fluid density, μ is dynamic viscosity, $\mathbf{g}$ is gravitational acceleration, and $\mathbf{K}$ is the (generally symmetric, positive-definite) permeability tensor; in an isotropic medium this reduces to a scalar permeability k times the identity. The presence of ∇p emphasizes that it is the *spatial variation* of pressure that drives flow.

A common alternative is hydraulic head

$$h = \frac{p}{\rho g} + z$$

(with $\mathbf{g} = -g\hat{k}$), in which case Darcy's law becomes

$$\mathbf{q} = -\mathbf{K}_h \nabla h$$

$$\mathbf{K}_h \equiv \frac{\rho g}{\mu} \mathbf{K}$$

where $\mathbf{K}_h$ is the hydraulic conductivity tensor.

Mass conservation introduces the divergence of the flux. For single-phase, incompressible flow with constant density and porosity, the continuity equation is

$$\nabla \cdot \mathbf{q} = Q$$

where $Q(\mathbf{x})$ represents volumetric sources ($Q > 0$) or sinks ($Q < 0$) per unit volume. Combining with Darcy's law yields an elliptic pressure (or head) equation:

Pressure form:

$$\nabla \cdot \left(\frac{1}{\mu} \mathbf{K} \nabla p \right) = \nabla \cdot \left(\frac{1}{\mu} \mathbf{K} \rho \mathbf{g} \right) - Q$$

If $\mathbf{K}$, μ , and $\rho \mathbf{g}$ are spatially constant and gravity is incorporated into a potential, this reduces to a Poisson equation. In an isotropic medium with constant k ,

$$\mu k \nabla^2 p = -Q$$

and in regions without sources or sinks ($Q = 0$) it further reduces to Laplace's equation,

$$\nabla^2 p = 0$$

Head form:

$$\nabla \cdot (\mathbf{K}_h \nabla h) = -Q$$

When compressibility or storage matters, time derivatives accompany these spatial operators. A common groundwater formulation uses hydraulic head and specific storage S_s :

$$S_s \frac{\partial h}{\partial t} = \nabla \cdot (\mathbf{K}_h \nabla h) + W$$

with W a volumetric source term per unit volume (positive for injection).

In all these expressions, the del operator ∇ is the unifying symbol: “grad” means ∇ acting on a scalar (e.g., ∇p or ∇h), while $\nabla \cdot$ and $\nabla \times$ denote divergence and curl acting on vector fields.